

Exercises

Write a python function for each of the following exercises:

A. Deadline: until seminar 2.

1. Calculate the **number** of rows in a matrix with the property that their elements are in increasing order.
2. Determine the **columns** of a matrix with the property that their smallest element is equal to 5.
3. Implement the bubble sort algorithm to sort **each row** of a matrix in ascending order.
4. Implement the insert sort algorithm to sort **each column** of a matrix in ascending order.
5. Write a recursive function to calculate the GCD (greatest common divisor) between two non-zero natural numbers.
6. Consider A and B two square matrices and n a non-zero natural number. Compute A^T , $A+B$, $A*B$ and A^n .
7. Implement the insertion sort algorithm for lists/vectors.
8. Verify the property of a permutation to be the identity permutation.
9. Let S be the set of binary vectors of length 7. Calculate, through random generation, a matrix A with 20 rows, vectors from S and a vector V with 20 elements, each $V[i]$ representing the quality of line i from A , defined as the sum of bits on line i .
10. Consider A and V built at exercise 9. Arrange the rows of matrix A such that the elements in V are in ascending order.

B. Deadline: until seminar 3.

11. Implement the hill-climbing algorithm to calculate the maximum of the function $f: \{1, 2, \dots, 2500\} \rightarrow \mathbb{R}, f(x) = (\sin(x - 2))^2 - x * \cos(x)$.
12. Consider $f: \{1, 2, \dots, 2500\} \rightarrow \mathbb{R}, f(x) = (\sin(x - 2))^2$ as the objective function of a maximization problem. Each phenotype $x \in \{1, 2, \dots, 2500\}$ corresponds to a genotype (binary string) obtained through the standard base-2 representation of x . Solve the maximization problem using a hill-climbing type algorithm.